

Uninet

DOMAIN
SOCIAL IMPACT

TEAM NAME

MIZU

SUBMISSION DATE

30 / 05 / 2026

01 TEAM INFORMATION

Provide the details of all team members below.

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02 PROBLEM STATEMENT

Missed Opportunities

Students frequently miss out on opportunities, such as societies and events, for which they are well-suited. Overall undergraduate participation in university societies and sports clubs—especially among female students—remains low. According to our surveys, female students are less motivated to participate because they lack a clear understanding of which activities exist or where they could showcase their talents.

Ineffective Communication

A common consensus among students is that society updates are typically shared via WhatsApp, and data is collected using Google Forms. In our observations, we saw that these methods are ineffective. Students experience "form fatigue" from repeatedly filling out Google Forms with the same data. Furthermore, they are not receiving sufficient information through these channels. Assuming these events are not for them, most students simply ignore the messages.

Additionally, current university communication relies heavily on generic notice boards, both physical and digital. The problem with this approach is that notifications are broad and do not target particular students. A student unfamiliar with the extracurricular landscape cannot easily determine which clubs align with their interests or understand the privileges associated with joining.

The Gap in the Market

Currently, there is no smart platform capable of analyzing an individual's preferences to promote relevant information about societies and events, nor is there a system to streamline the registration process and committee recruitment based on that collected data.

03 PROPOSED SOLUTION

This solution is a smart student engagement platform designed to involve undergraduates with university experience by introducing what are the most suitable societies for them. Powered by an intelligent matching engine, it analyzes individual skills, interests, and behaviors to provide highly personalized society and event recommendations. To ensure sustained engagement, the platform features a dynamic gamification system,

- rewarding students with digital badge system for their achievements.
- turning badges publicly visible for community encouragement and participation history.

3.1 Solution Overview

A personalized platform (web application) for students to scroll through. This platform is,

- designed to seamlessly connect students with society leaders, clubs and events that truly matter to them.
- specially built for universities according to the Sri Lankan module system

By analyzing each student's unique profile — their interests, skills, and academic background — it delivers targeted notifications and promotion flyers that make university life more meaningful and accessible from day one.

3.2 Key Features

1. Smart Student Profile

- Students create a **one-time profile** providing their:
 - Basic personal information
 - Interests, hobbies, and skills
 - Academic information (Course modules)
 - Availability and preferences
- This eliminates **form fatigue** — data is entered once and reused automatically across all other registrations

2. Personalized Recommendation Engine

- An AI/ML-based intelligent engine analyzes the student's profile and recommends:
 - Relevant **societies and clubs**
 - Upcoming **events and competitions**
 - **Committee roles** they may be well-suited for
- Recommendations are tailored, not generic — students can easily see what's **most relevant** to them.

3. Targeted Notification System

- Replaces WhatsApp broadcasts and generic notice boards with **smart, personalized alerts**.
- Notifications are affected by student interests and past engagements.
- Reduces message overload and prevents students from ignoring updates

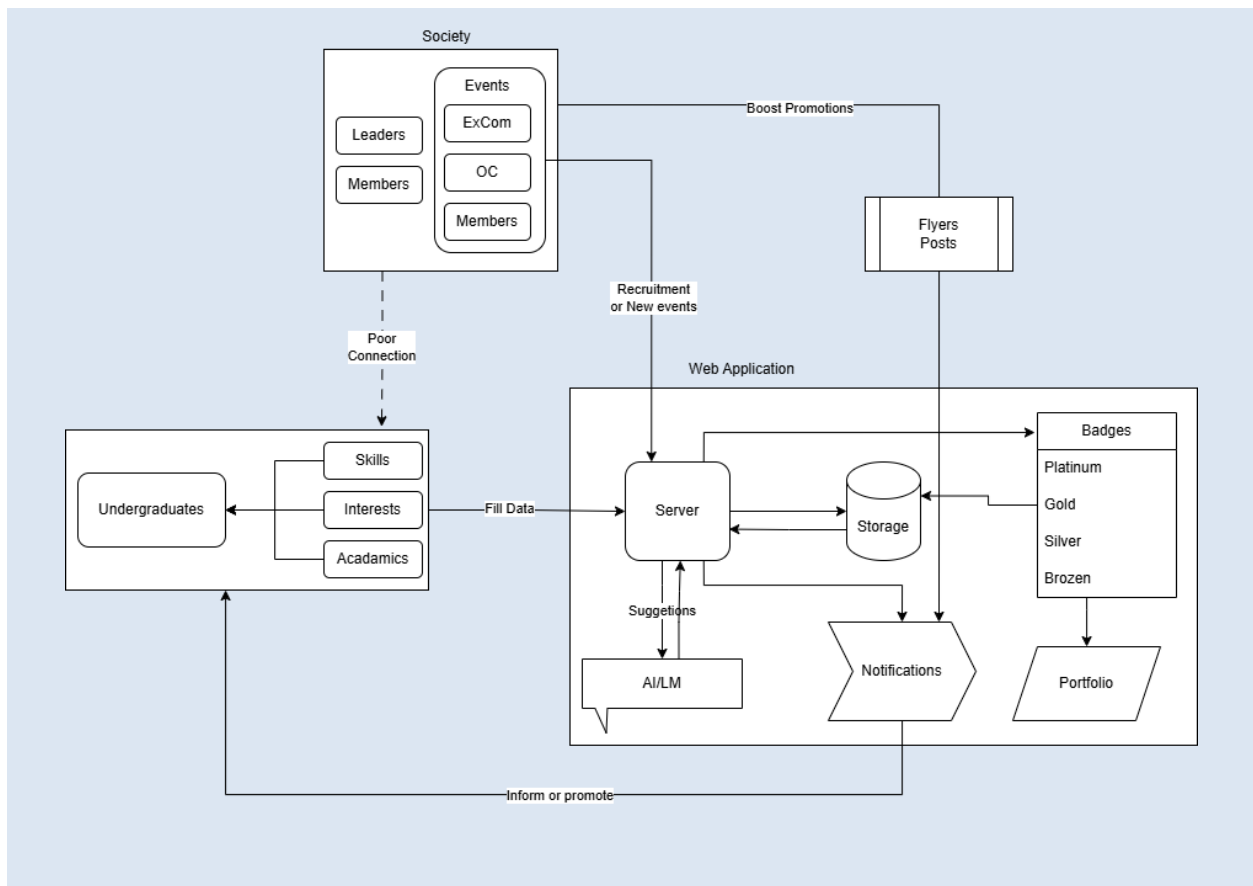
4. Streamlined Registration & Recruitment

- **One-click registration** for events using pre-filled profile data.
- Automated **committee recruitment pipeline** that matches students to roles based on their skills and interests.
- Tracks participation history to build a student's **co-curricular portfolio**

5. Society & Admin Dashboard

- Society committees can:
 - Post events and opportunities
 - View **matched student suggestions** for recruitment
 - Track registrations and engagement analytics
- Society admins can monitor **overall participation trends** and identify underrepresented groups.

3.3 Architecture / System Diagram



04 TECHNOLOGY STACK

Component	Technology / Tool	Purpose
Frontend	React Native	Provides a seamless - cross-platform mobile interface - easily browse societies - view personalized recommendations - track badges and achievements ongoing.
Backend	Python (Flask / FastAPI)	Centralized brain of the application. Using a lightweight, Python-based framework (like Flask) allows for rapid development of API routing and seamlessly integrates with the AI models natively. It securely handles user authentication and database operations.
Database	PostgreSQL + Firebase	PostgreSQL reliably stores structured relational data, - student profiles - society details - event schedules Firebase enables real-time features like, - instant chat - live updates - dynamic badge unlocking.
Recommendation	Python (scikit-learn / collaborative filtering)	Powers the core "smart matching engine." It analyzes individual student interests and past behaviors to generate, - the tailored society and event suggestions - eliminating the need for generic notice boards.
Notifications	Push notifications + In-app alerts	Replaces ineffective WhatsApp groups by driving continuous, targeted engagement. It delivers personalized alerts for relevant upcoming events and rewards directly to the student's device.

05 INNOVATION & ORIGINALITY

5.1 Competitive Advantage

Existing Solutions	Our Solution
Limited: Reliance on fragmented WhatsApp groups and generic physical/digital notice boards, failing to target specific student interests.	Smart: An AI-powered smart matching engine that analyzes individual profiles (skills, interests) to deliver highly personalized society and event recommendations.
Inefficient: Data collection and event registration rely on repetitive Google Forms, causing "form fatigue" and high-friction onboarding.	Targeted: A centralized platform that streamlines the registration process, utilizing a single, unified student profile to completely eliminate repetitive data entry.
General: Lack of ongoing motivation or systems to track a student's extracurricular growth and achievements over their university lifecycle.	Encourage: A dynamic gamification system featuring badges and rewards that turns everyday community participation into a visible, trackable achievement journey.

06 REAL-WORLD IMPACT & SCALABILITY

6.1 Target Audience

- **Primary Users:** Undergraduate students, particularly first-years and female students who currently experience a disconnect in campus engagement and need personalized guidance to find where they belong.
- **Secondary Users:** Society leaders, organizing committees (such as IEEE student branches, tech clubs, or sports captains) and event coordinators who need efficient, targeted recruitment tools rather than shouting into the void of generic notice boards.

6.2 Expected Impact

- **Bridging the Engagement Gap:** By delivering personalized recommendations, the platform directly lowers the barrier to entry, specifically encouraging underrepresented demographics (like female undergraduates) to actively participate in spaces where their talents can shine.
- **Eliminating Administrative Friction:** Eradicating the "form fatigue" of repetitive Google Forms and the noise of scattered WhatsApp groups saves hundreds of hours for both students and organizing committees.
- **Tangible Skill Development:** The gamified badge system turns abstract participation into a visible, trackable portfolio. Students build verifiable records of their soft skills, teamwork, and leadership, which becomes highly valuable when presenting themselves at university job fairs and to future employers.

6.3 Scalability Plan

- **Phase 1: Pilot & Validation (Months 0–6)** Launch an initial pilot at Wayamba University of Sri Lanka to validate the core matchmaking engine and user experience. Onboard a select group of highly active early adopters, such as the local IEEE Computer Society Student Branch Chapter and major sports clubs, to test the system, gather feedback, and refine the recommendation algorithms.
 - **Phase 2: Cross-Campus Expansion (Months 6–12)** Package the platform as a scalable Software-as-a-Service (SaaS) or white-label solution. Roll it out to other state and private universities, adapting the database structure to handle multiple independent university ecosystems while maintaining centralized cloud architecture.
 - **Phase 3: Ecosystem Integration (Year 1+)** Expand the platform's utility beyond just societies by integrating it with existing university data sets that students gathered locally. Additionally, open a module for corporate partners and recruiters, allowing students to seamlessly export their earned society badges and event participation data as verified extracurricular resumes.
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07 MARKET VIABILITY

1. **Market Demand & Validation** The demand for a centralized engagement platform is universally high across university campuses. With dozens of active societies and thousands of students struggling with generic notice boards and "form fatigue," the immediate market is a captive audience of undergraduates actively seeking a smoother way to connect. Because the platform solves a dual-sided problem,
 - helping students find clubs
 - helping organizing committees find the right talentsadoption will be driven by mutual necessity.
2. **Phase 1: Securing Initial Seed Funding** To cover initial development, server costs, and the database infrastructure, the platform will rely on a community-driven seed strategy:

- **Corporate Sponsorships (Talent Pipeline):** Companies are always looking for proactive student leaders. By tapping into professional networks and relationships built during university job fairs, we can offer early corporate sponsors premium visibility or branding on the platform in exchange for seed funding.
 - **Community Grants & Hackathons:** Leveraging technical communities and established student branches (such as IEEE initiatives or university tech grants) to secure micro-grants dedicated to campus improvement projects.
 - **Hackathon Prize Capital:** Reinvesting any initial hackathon winnings directly into the platform's first-year hosting and operational costs.
- 3. Phase 2: Student-Funded Sustainability Model** Once the initial funding phase successfully launches the platform, the financial model will transition to a self-sustaining, student-funded ecosystem:
- **The "Buy-a-Coffee" Crowdfunding Model:** A transparent, community-supported approach where students can make micro-donations to keep the platform running, similar to open-source or community-driven projects.
 - **Premium Society Features:** The platform remains 100% free for individual students. However, organizing committees or clubs can pool small amounts of student funds to unlock "Premium Society Tools," such as advanced recruitment analytics, bulk event messaging, or priority listing on the recommendation engine.
 - **Alumni "Pay-It-Forward" Sponsorships:** Graduated students who benefited from the platform to build their portfolios can sponsor server costs for a month, earning a permanent "Platform Benefactor" badge on their alumni profile.

DECLARATION

We, the undersigned, declare that this proposal is our original work and has not been submitted elsewhere. All information is accurate to the best of our knowledge.

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Date: 30/5/2026

HackElite 3.0

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